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Age at Diagnosis and Clinical and Genetic Profiles Among Adults With Attention-Deficit/Hyperactivity Disorder

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Abstract

IMPORTANCE Adult attention-deficit/hyperactivity disorder (ADHD) diagnoses are rising, yet distinctions between those diagnosed in childhood or adolescence vs adulthood remain underexplored.

OBJECTIVES To characterize differences in demographic, clinical, and genetic profiles between adults with ADHD diagnosed in childhood or adolescence (youth-diagnosed) vs adulthood (adult-diagnosed), and to compare the profile of each group with that from matched individuals without ADHD.

DESIGN, SETTING, AND PARTICIPANTS This cross-sectional, multicenter study used data collected in the All of Us Research Program, a nationwide US cohort integrating clinical and genetic data, from June 2017 to September 2023. Analyses were performed between July and December 2025. Participants were adults who self-reported an ADHD diagnosis, dichotomized by age at first diagnosis into youth-diagnosed (0-17 years) and adult-diagnosed (≥18 years) ADHD groups. Each ADHD group was independently matched 1:2 via propensity scores to controls without ADHD on age, sex, race and ethnicity, educational attainment, and income.

MAIN OUTCOMES AND MEASURES The primary outcomes were self-reported psychiatric or somatic co-occurring conditions, measures of functional impairment (general health and daily functioning), and ADHD polygenic risk scores.

RESULTS Of 13 527 adults with ADHD, 3853 (28.5%; mean [SD] age, 34.8 [11.5] years; 71.5% female) were youth-diagnosed and 9674 (71.5%; mean [SD] age, 43.8 [14.3] years; 75.3% female) adult-diagnosed. Adult-diagnosed individuals were older and had higher educational attainment (eg, 548 [14.2%] vs 2690 [27.8%] for advanced degree) and annual household income (eg, 364 [9.4%] vs 1407 [14.5%] for $100 001-$150 000). Adjusting for age and sex, adult- versus youth-diagnosed ADHD was associated with lower adjusted probabilities of most psychiatric comorbidities (8 of 9 conditions) with the largest absolute differences for posttraumatic stress disorder (−7.50 [95% CI, −9.25 to −5.88] percentage points [pp]), bipolar disorder (−7.44 [95% CI, −8.99 to −6.00] pp), and personality disorder (−4.78 [95% CI, −5.89 to −3.81] pp). For somatic conditions (6 of 9 categories), the largest absolute differences were for lung conditions (−7.88 [95% CI, −9.73 to −5.79] pp), digestive conditions (−7.27 [95% CI, −9.12 to −5.28] pp), and bone, joint, and muscle conditions (−7.00 [95% CI, −8.78 to −5.20] pp). For impairment in overall health and daily functioning (7 of 9 measures), the largest absolute differences were for physical health (−3.63 [95% CI, −4.83 to −2.44] pp), being bothered by emotional problems in the past 7 days (−2.51 [95% CI, −3.79 to −1.24] pp), and general health (−2.26 [95% CI, −3.40 to −1.18] pp). Compared with their matched controls without ADHD, both ADHD groups showed markedly higher probability differences of comorbid psychiatric (eg, for depression, 39.33 [95% CI, 38.19-40.48] pp for adult-diagnosed ADHD and 36.92 [95% CI, 35.05-38.83] pp for youth-diagnosed ADHD) and

(continued)

Key Points

Question Do demographic, clinical, and genetic profiles of adults diagnosed with attention-deficit/hyperactivity disorder (ADHD) in adulthood differ from adults diagnosed with ADHD in childhood or adolescence, and how do profiles of both groups compare with matched controls without ADHD?

Findings In this cross-sectional study of 13 527 adults with ADHD, adult-diagnosed individuals had significantly fewer psychiatric and somatic comorbidities, less impairment in overall health and daily activities, and lower ADHD polygenic risk scores than youth-diagnosed individuals. Yet both ADHD groups had significantly higher psychiatric and somatic comorbidities, overall functional impairments, and ADHD polygenic risk scores than matched controls.

Meaning These findings suggest that adult-diagnosed ADHD may represent a comparatively milder clinical and genetic profile than youth-diagnosed ADHD, yet both groups showed substantial clinical burden.

+ Supplemental content

Author affiliations and article information are listed at the end of this article.

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Abstract (continued)

somatic (eg, for brain and nervous system conditions, 21.86 [95% CI, 20.64-22.93] pp for adult-diagnosed ADHD and 22.84 [95% CI, 21.05-24.83] pp for youth-diagnosed ADHD) conditions, and functional impairments (eg, for bothered by emotional problems past 7 days, 6.11 [95% CI, 5.45-6.80] pp for adult-diagnosed ADHD and 7.29 [95% CI, 5.99-8.60] pp for youth-diagnosed ADHD). ADHD polygenic risk scores were lower in adult- vs youth-diagnosed groups (Cohen *d*, −0.17 [95% CI, −0.21 to −0.12]) and higher in youth-diagnosed (Cohen *d*, 0.21 [95% CI, 0.16-0.26]) and adult-diagnosed (Cohen *d*, 0.12 [95% CI, 0.09-0.15]) groups compared with controls.

CONCLUSIONS AND RELEVANCE In this cross-sectional study of adults with ADHD, individuals first diagnosed with ADHD in adulthood showed a milder ADHD-like clinical and genetic profile than those diagnosed in childhood or adolescence, which may contribute to later recognition of the disorder. Nonetheless, adults diagnosed either as youths or adults exhibited profiles characteristic of ADHD relative to matched controls without ADHD, including substantial burden, supporting adult-diagnosed ADHD as a clinically important subgroup warranting recognition and management.

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Introduction

The number of adults diagnosed with attention-deficit/hyperactivity disorder (ADHD) has increased over time.[1] It is estimated that over half of adults with ADHD in the US were first diagnosed in adulthood,[2] and this pattern has coincided with a parallel increase in prescriptions of stimulant medications, with the most pronounced increases observed among adults, particularly adult women.[3] Whether the increases in adult ADHD diagnosis reflect increased recognition of true cases, a true increase in ADHD symptoms at the population level, misdiagnosis,[4] secondary gain,[5] or stimulant misuse[6] remains a topic of active debate.[7]

Despite these uncertainties, studies comparing the clinical profiles of adults diagnosed with ADHD in childhood with individuals first diagnosed in adulthood are scarce and largely limited to small clinic-based samples.[8,9] In a single-center retrospective medical record review, adults diagnosed in childhood had a higher overall burden of comorbidity (notably depression) than those first diagnosed in adulthood,[8] whereas a specialty adult ADHD clinic sample reported broadly similar symptom, functioning, and quality-of-life profiles across groups.[9]

To address this gap, we leveraged data from the All of Us Research Program (All of Us) to compare adults with ADHD diagnosed in childhood or adolescence (youth-diagnosed) and those diagnosed in adulthood (adult-diagnosed) across demographic characteristics, co-occurring psychiatric and somatic conditions, self-reported health and functional impairments, and genetic liability for ADHD as measured by ADHD polygenic risk scores (ADHD-PRSs). Additionally, we compared each ADHD group to a matched control group of individuals without ADHD. Exploratory analyses examined whether findings differed by sex and whether separating youth-diagnosed ADHD into childhood- and adolescence-diagnosed groups revealed distinct outcome patterns. We hypothesized that adults diagnosed with ADHD in adulthood would not differ in their clinical and genetic profiles from those diagnosed as youth, and that both ADHD groups would exhibit profiles distinct from adults without ADHD.

Methods

Participants

This cross-sectional study used data from the All of Us Research Program, a large observational study aiming to enroll participants throughout the US to provide insight into interindividual differences in

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health outcomes.[10] Additional details on All of Us recruitment and enrollment procedures are provided in the eMethods in Supplement 1. Institutional review boards at each participating site approved the program, and all participants provided written informed consent. The present analysis used deidentified data available through the All of Us Researcher Workbench. This study followed the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) reporting guideline for cross-sectional studies.

We analyzed data from participants in the All of Us (Controlled Tier Dataset version 8 [C2024Q3R5]) who completed the Personal and Family Health History questionnaire (N = 298 091). The survey responses included in this analysis were collected between June 2017 and September 2023.

Outcomes

Participants were classified as having ADHD based on affirmative responses to the question "Are you still seeing a doctor or health care provider for ADHD?" from the Personal and Family Health History questionnaire. To further categorize individuals based on age of ADHD diagnosis, we used their responses to the question "About how old were you when you were first told you had ADHD?" Participants answering a child (0-11 years) or adolescent (12-17 years) were classified as youth-diagnosed ADHD, while those answering an adult (18-64 years), older adult (65-74 years), or older adult (≥75 years) were classified as adult-diagnosed ADHD. Individuals missing the age at diagnosis were excluded.

Psychiatric conditions were ascertained from the same survey using questions following an identical wording pattern "Are you still seeing a doctor or health care provider for [condition]?" Conditions assessed included anxiety reaction or panic disorder, social phobia, autism spectrum disorder, bipolar disorder, depression, eating disorder, personality disorder, posttraumatic stress disorder, schizophrenia, alcohol use disorder, and drug use disorder. For this study, anxiety reaction or panic disorder and social phobia were grouped together under anxiety disorder, and alcohol use disorder and drug use disorder were grouped as substance use disorder.

Somatic conditions were also ascertained from the Personal and Family Health History questionnaire and comprised 9 categories of conditions: bone, joint, and muscle; brain and nervous system; cancer; digestive; hearing and eye; heart and blood; hormone and endocrine; kidney; and lung. A category was coded present when the respondent endorsed seeing a clinician for at least 1 condition within that category. Description of conditions within categories is provided in eTable 1 in Supplement 1.

Impairment was measured using self-reported ratings of general health and daily functioning, including general health, physical health, mental health, quality of life, satisfaction with social activities, ability to perform physical and social roles, fatigue, and emotional distress. Further details are provided in the eMethods in Supplement 1.

Covariates

Demographic and socioeconomic variables were self-reported from the All of Us survey The Basics. Sex at birth was reported as female, male, or neither (included intersex, none of these, prefer not to answer, and skip). Self-reported race and ethnicity were assessed to more fully characterize the cohort and coded as Hispanic, non-Hispanic Black or African American, non-Hispanic White, other (included American Indian or Alaska Native, Asian, Native Hawaiian or Other Pacific Islander, Middle Eastern or North African, and more than 1 race or ethnicity), or not reported (included none of these, prefer not to answer, and skip). Educational achievement was categorized as less than high school, high school graduate or General Educational Development, some college, college graduate, advanced degree, or not reported. Annual household income was analyzed in ordered categories with an additional not-reported level.

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Propensity Score Matching

For each ADHD group (youth-diagnosed and adult-diagnosed), we constructed an independent control cohort using 1:2 nearest-neighbor propensity score matching without replacement.[11] The propensity score was estimated using logistic regression as the predicted probability of ADHD conditional on age at survey, sex, race and ethnicity, educational achievement, and household income. "Not reported" levels were retained as categories where applicable. Eligible controls were participants who reported no lifetime ADHD. Matching was implemented separately within each group, with complete-case inclusion for the matching covariates. Covariate balance was evaluated by standardized mean differences, with absolute values less than 0.10 considered acceptable. In a sensitivity analysis, we repeated the 1:2 nearest-neighbor matching to construct independent clinical control cohorts (matched clinical controls), restricting eligible controls to participants who reported seeing a clinician for at least 1 somatic condition. Propensity scores were re-estimated using the same covariates. These analyses evaluated whether the main findings were robust when using a clinically ascertained control sample.

PRS

ADHD-PRSs were calculated from All of Us Controlled Tier array genotypes[12] using PRSice, version 2.3.5,[13] and summary statistics from the most recent ADHD genome-wide association study.[14] After standard sample- and variant-level quality control,[15] PRSs were computed across 11 genome-wide association study *P* value thresholds, and the first principal component across thresholds was used as the ADHD-PRS measure.[16] Additional details are provided in the eMethods in Supplement 1.

Statistical Analysis

For adult- vs youth-diagnosed ADHD, we fit multivariable logistic regression models for each psychiatric condition, somatic category, and impairment outcome, adjusting for age and sex. In exploratory analyses, we separated the youth-diagnosed group into child-diagnosed (0-11 years of age) and adolescent-diagnosed (12-17 years of age) groups and repeated pairwise comparisons among adult-, adolescent-, and child-diagnosed ADHD. Sex moderation was examined by including an interaction term between diagnosis timing and sex. We report predicted probabilities and odds ratios (ORs) with 95% CIs. As sensitivity analyses, we repeated these comparisons with additional adjustment for educational achievement, and separately with additional adjustment for income. Educational achievement and income were examined in separate sensitivity models to reduce collinearity concerns and preserve model interpretability. Each ADHD group was also compared with its own propensity score–matched control cohort using logistic regression without additional covariates.

For genetic analyses, ADHD-PRSs were regressed on age, sex, and the first 10 genetic ancestry principal components in the full genetic subsample. Standardized residuals from this model were used in all PRS analyses. Group differences in PRSs were estimated with linear regression and summarized as mean differences with 95% CIs and the Cohen *d* with 95% CIs. In addition, we performed quintile analyses of the residualized ADHD-PRS. Global cut points were defined in the pooled analytic sample using the 20th, 40th, 60th, and 80th percentiles of the standardized residuals, ensuring identical thresholds across contrasts. Logistic regression models estimated ORs and 95% CIs for quintiles 2 through 5 compared with 1 (reference).

All tests were 2-sided. The false discovery rate (FDR) was controlled at .05 across outcomes using the Benjamini-Hochberg procedure, and $P < .05$ was considered statistically significant. Analyses were conducted between July and December 2025 in R, version 4.4.0 (R Project for Statistical Computing), within All of Us Researcher Workbench.

Results

Of 298 091 participants who completed the Personal and Family Health History questionnaire, 13 527 adults with ADHD were identified (10 035 [74.2%] female, 3414 [25.2%] male). Most respondents

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self-reported race and ethnicity as Hispanic (1329 [9.8%]), non-Hispanic Black or African American (497 [3.7%]), or non-Hispanic White (10 155 [75.1%]). Of 13 527 adults with ADHD, 3853 (28.5%) were youth-diagnosed individuals (mean [SD] age, 34.8 [11.5] years; 2754 [71.5%] female) and 9674 (71.5%) were adult-diagnosed individuals (mean [SD] age, 43.8 [14.3] years; 75.3% female) (**Table 1**). Compared with the youth-diagnosed group, adult-diagnosed individuals had higher educational attainment (eg, 548 [14.2%] vs 2690 [27.8%] for advanced degree) and household income (eg, 364 [9.4%] vs 1407 [14.5%] for $100 001-$150 000).

Individuals with adult-diagnosed ADHD, compared with youth-diagnosed ADHD, exhibited lower adjusted probabilities of anxiety disorder (adjusted difference, −3.08 [95% CI, −5.10 to −1.12] percentage points [pp]), autism spectrum disorder (−2.36 [95% CI, −3.19 to −1.60] pp), bipolar disorder (−7.44, [95% CI, −8.99 to −6.00] pp), eating disorder (−0.89 [95% CI, −1.82 to −0.04] pp), personality disorder (−4.78 [95% CI, −5.89 to −3.81] pp), posttraumatic stress disorder (−7.50 [95% CI, −9.25 to −5.88] pp), schizophrenia (−1.24 [95% CI, −1.83 to −0.73] pp), and substance use disorder (−3.65 [95% CI, −4.65 to −2.57] pp), but not depression (−0.97 [95% CI, −2.94 to 1.04] pp) (**Figure 1**A). Similar results were observed in sensitivity analyses further adjusted for educational achievement or income, with the exception of anxiety and eating disorders, which no longer differed between groups (**Table 2**).

Individuals with adult-diagnosed ADHD were less likely to report somatic conditions compared with youth-diagnosed individuals within the following categories: bone, joint, and muscle (−7.00

Table 1. Characteristics of Adults With ADHD by Age at First Diagnosis

Characteristic	Participants, No. (%)		
	Youth-diagnosed ADHD (n = 3853)	Adult-diagnosed ADHD (n = 9674)	SMD[a]
Age, mean (SD), y	34.8 (11.5)	43.8 (14.3)	0.70
Sex			
Female	2754 (71.5)	7281 (75.3)	0.09
Male	1077 (28.0)	2337 (24.2)	0.09
Neither selected	22 (0.6)	56 (0.6)	<0.01
Race and ethnicity[b]			
Hispanic	383 (9.9)	946 (9.8)	<0.01
Non-Hispanic Black	131 (3.4)	366 (3.8)	0.02
Non-Hispanic White	2839 (73.7)	7316 (75.6)	0.04
Not reported	95 (2.5)	182 (1.9)	0.04
Other[c]	405 (10.5)	864 (8.9)	0.05
Educational achievement			
<high school	143 (3.7)	121 (1.3)	0.16
High school or GED	615 (16.0)	679 (7.0)	0.28
Some college	1481 (38.4)	2889 (29.9)	0.18
College graduate	1002 (26.0)	3182 (32.9)	0.15
Advanced degree	548 (14.2)	2690 (27.8)	0.33
Not reported	64 (1.7)	113 (1.2)	0.04
Annual household income, $			
<10 000	555 (14.4)	653 (6.8)	0.25
10 00-24 999	566 (14.7)	1139 (11.8)	0.09
25 000-34 999	358 (9.3)	730 (7.5)	0.06
35 000-49 999	420 (10.9)	1055 (10.9)	0.00
50 000-74 999	489 (12.7)	1427 (14.8)	0.06
75 000-99 999	319 (8.3)	1188 (12.3)	0.13
100 000-149 999	364 (9.4)	1407 (14.5)	0.16
150 000-199 999	171 (4.4)	659 (6.8)	0.10
≥200 000	193 (5.0)	770 (8.0)	0.12
Not reported	418 (10.8)	646 (6.7)	0.15

Abbreviations: ADHD, attention-deficit/hyperactivity disorder; GED, General Educational Development; SMD, standardized mean difference.

[a] SMD 0.10 or higher was interpreted as a meaningful imbalance.

[b] Race and ethnicity was self-reported based on All of Us survey classifications.

[c] Other included American Indian or Alaska Native, Asian, Middle Eastern or North African, Native Hawaiian or Other Pacific Islander, and more than 1 race or ethnicity.

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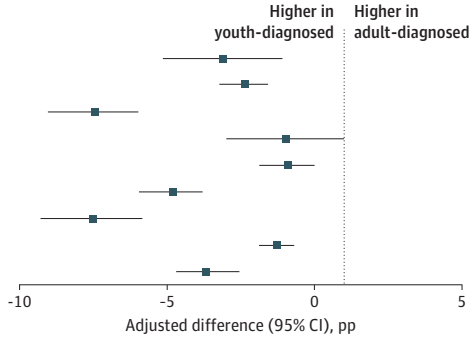
[95% CI, −8.78 to −5.20] pp), brain and nervous system (−6.64 [95% CI, −8.51 to −4.48] pp), digestive (−7.27 [95% CI, −9.12 to −5.28] pp), heart and blood (−5.29 [95% CI, −7.12 to −3.46] pp), kidney (−1.64 [95% CI, −2.76 to −0.56] pp), and lung (−7.88 [95% CI, −9.73 to −5.79] pp) (Figure 1B). No statistically significant adjusted differences were observed for cancer (−0.71 [95% CI, −1.87 to 0.37] pp), hearing and eye (0.40 [95% CI, −1.40 to 2.01] pp), and hormone and endocrine (−1.77 [95% CI, −3.64 to −0.09] pp) conditions. Similar results were observed in sensitivity analyses further adjusted for education or income, with the exception of kidney conditions, which no longer differed between groups (Table 2).

The adult-diagnosed ADHD group reported lower impairment across 7 of 9 measures of health and daily functioning compared with the youth-diagnosed group (eg, for bothered by emotional problems past 7 days adjusted for sex and age: probability difference, −2.51 [95% CI, −3.79 to −1.24] pp) (Figure 1C). After adjusting for sex, age, and educational achievement (eg, for bothered by emotional problems past 7 days: probability difference, −0.49 [95% CI, −1.85 to 0.68] pp) or for sex,

Figure 1. Dot Plots of Adjusted Probability Differences of Co-Occurring Conditions and Measures of Impairment Between Adult- and Youth-Diagnosed Attention-Deficit/Hyperactivity Disorder (ADHD)

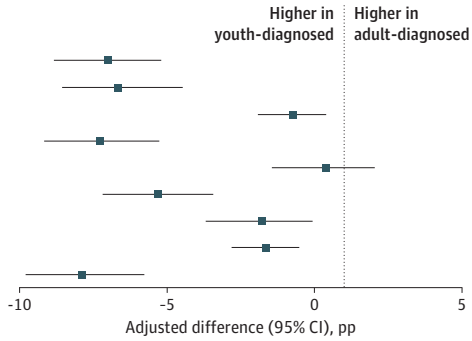
A Psychiatric comorbidities

Comorbidities	Adjusted difference (95% CI), pp
Anxiety disorder	−3.08 (−5.10 to −1.12)
Autism spectrum disorder	−2.36 (−3.19 to −1.60)
Bipolar disorder	−7.44 (−8.99 to −6.00)
Depression	−0.97 (−2.94 to 1.04)
Eating disorder	−0.89 (−1.82 to −0.04)
Personality disorder	−4.78 (−5.89 to −3.81)
Posttraumatic stress disorder	−7.50 (−9.25 to −5.88)
Schizophrenia	−1.24 (−1.83 to −0.73)
Substance use disorder	−3.65 (−4.65 to −2.57)



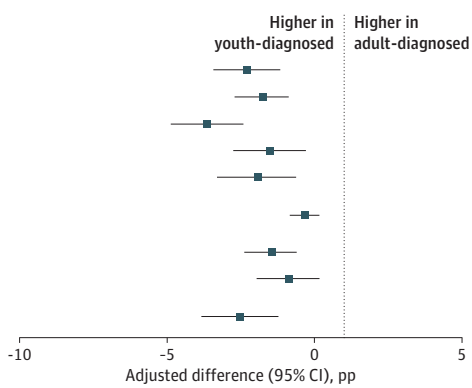
B Somatic comorbidity condition

Comorbidities	Adjusted difference (95% CI), pp
Bone, joint, and muscle	−7.00 (−8.78 to −5.20)
Brain and nervous system	−6.64 (−8.51 to −4.48)
Cancer	−0.71 (−1.87 to 0.37)
Digestive	−7.27 (−9.12 to −5.28)
Hearing and eye	0.40 (−1.40 to 2.01)
Heart and blood	−5.29 (−7.12 to −3.46)
Hormone and endocrine	−1.77 (−3.64 to −0.09)
Kidney	−1.64 (−2.76 to −0.56)
Lung	−7.88 (−9.73 to −5.79)



C Overall health and daily activities

Comorbidities	Adjusted difference (95% CI), pp
General health	−2.26 (−3.40 to −1.18)
Quality of life	−1.74 (−2.66 to −0.90)
Physical health	−3.63 (−4.83 to −2.44)
Mental health	−1.49 (−2.72 to −0.32)
Satisfaction with social activities	−1.92 (−3.25 to −0.65)
Ability to carry out everyday physical activities	−0.31 (−0.79 to 0.08)
Average fatigue in the past 7 days	−1.44 (−2.33 to −0.65)
Ability to carry out usual social roles	−0.87 (−1.92 to 0.15)
Bothered by emotional problems past 7 days	−2.51 (−3.79 to −1.24)



Dot-and-whisker plots show adjusted probability differences of adult-diagnosed ADHD minus youth-diagnosed ADHD, in percentage points (pp). A false discovery rate of less than .05 was applied across 3 outcome categories, each with a total of 9 tests.

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Table 2. Adjusted Probability Differences and ORs Comparing Adult-Diagnosed ADHD With Youth-Diagnosed ADHD

Diagnosis or impairment	Adjusted for sex and age			Adjusted for sex, age, and educational attainment			Adjusted for sex, age, and income		
	Probability difference (95% CI), pp[a]	AOR (95% CI)[b]	*P* value[c]	Probability difference (95% CI), pp[a]	AOR (95% CI)[b]	*P* value[c]	Probability difference (95% CI), pp[a]	AOR (95% CI)[b]	*P* value[c]
Psychiatric comorbidities									
Anxiety disorder	−3.08 (−5.10 to −1.12)	0.88 (0.81 to 0.95)	.002	−0.28 (−2.27 to 1.65)	0.99 (0.91 to 1.07)	.78	−0.84 (−2.64 to 1.11)	0.96 (0.89 to 1.05)	.39
Autism spectrum disorder	−2.36 (−3.19 to −1.60)	0.57 (0.48 to 0.69)	<.001	−1.98 (−2.88 to −1.22)	0.62 (0.52 to 0.75)	<.001	−2.02 (−2.90 to −1.23)	0.62 (0.52 to 0.74)	<.001
Bipolar disorder	−7.44 (−8.99 to −6.00)	0.58 (0.52 to 0.64)	<.001	−3.97 (−5.37 to −2.59)	0.73 (0.65 to 0.81)	<.001	−4.55 (−6.01 to −3.13)	0.69 (0.62 to 0.77)	<.001
Depression	−0.97 (−2.94 to 1.04)	0.96 (0.88 to 1.04)	.31	1.68 (−0.35 to 3.63)	1.08 (0.99 to 1.17)	.11	1.23 (−0.73 to 3.11)	1.06 (0.97 to 1.15)	.26
Eating disorder	−0.89 (−1.82 to −0.04)	0.84 (0.71 to 0.99)	.04	−0.49 (−1.39 to 0.34)	0.91 (0.76 to 1.08)	.30	−0.39 (−1.31 to 0.47)	0.92 (0.78 to 1.10)	.39
Personality disorder	−4.78 (−5.89 to −3.81)	0.46 (0.39 to 0.53)	<.001	−3.00 (−4.02 to −2.05)	0.59 (0.50 to 0.69)	<.001	−3.19 (−4.14 to −2.29)	0.57 (0.49 to 0.66)	<.001
Posttraumatic stress disorder	−7.50 (−9.25 to −5.88)	0.69 (0.63 to 0.75)	<.001	−4.08 (−5.98 to −2.29)	0.81 (0.74 to 0.89)	<.001	−4.10 (−5.82 to −2.54)	0.80 (0.73 to 0.88)	<.001
Schizophrenia	−1.24 (−1.83 to −0.73)	0.44 (0.32 to 0.60)	<.001	−0.52 (−1.01 to −0.08)	0.68 (0.49 to 0.94)	.03	−0.66 (−1.18 to −0.24)	0.62 (0.45 to 0.85)	.004
Substance use disorder	−3.65 (−4.65 to −2.57)	0.53 (0.45 to 0.62)	<.001	−2.05 (−3.02 to −1.18)	0.68 (0.57 to 0.80)	<.001	−2.20 (−3.20 to −1.26)	0.66 (0.56 to 0.78)	<.001
Somatic comorbidity condition									
Bone, joint, and muscle	−7.00 (−8.78 to −5.20)	0.71 (0.65 to 0.78)	<.001	−5.04 (−6.84 to −3.27)	0.78 (0.71 to 0.85)	<.001	−4.73 (−6.43 to −2.87)	0.79 (0.72 to 0.86)	<.001
Brain and nervous system	−6.64 (−8.51 to −4.48)	0.76 (0.70 to 0.82)	<.001	−4.43 (−6.52 to −2.62)	0.83 (0.77 to 0.90)	<.001	−4.59 (−6.39 to −2.70)	0.82 (0.76 to 0.89)	<.001
Cancer	−0.71 (−1.87 to 0.37)	0.89 (0.74 to 1.07)	.22	−0.73 (−1.90 to 0.42)	0.88 (0.73 to 1.06)	.25	−0.76 (−1.81 to 0.35)	0.88 (0.73 to 1.06)	.22
Digestive	−7.27 (−9.12 to −5.28)	0.73 (0.67 to 0.79)	<.001	−5.03 (−6.93 to −3.03)	0.80 (0.74 to 0.87)	<.001	−5.55 (−7.39 to −3.81)	0.78 (0.72 to 0.85)	<.001
Hearing and eye	0.40 (−1.40 to 2.01)	1.02 (0.94 to 1.11)	.66	−0.91 (−2.53 to 0.94)	0.96 (0.88 to 1.04)	.32	−0.36 (−1.99 to 1.40)	0.98 (0.90 to 1.07)	.69
Heart and blood	−5.29 (−7.12 to −3.46)	0.78 (0.72 to 0.85)	<.001	−3.78 (−5.66 to −2.03)	0.84 (0.77 to 0.91)	<.001	−3.81 (−5.68 to −1.97)	0.83 (0.77 to 0.91)	<.001
Hormone and endocrine	−1.77 (−3.64 to −0.09)	0.91 (0.83 to 1.00)	.06	−0.95 (−2.64 to 0.78)	0.95 (0.87 to 1.05)	.32	−1.08 (−2.78 to 0.63)	0.94 (0.86 to 1.04)	.26
Kidney	−1.64 (−2.76 to −0.56)	0.75 (0.63 to 0.90)	.002	−1.11 (−2.25 to −0.01)	0.82 (0.69 to 0.98)	.04	−1.07 (−2.12 to −0.03)	0.83 (0.69 to 0.99)	.05
Lung	−7.88 (−9.73 to −5.79)	0.70 (0.64 to 0.76)	<.001	−6.31 (−8.08 to −4.41)	0.75 (0.68 to 0.81)	<.001	−6.15 (−8.05 to −4.27)	0.75 (0.69 to 0.82)	<.001
Overall health and daily activities									
General health	−2.26 (−3.40 to −1.18)	0.71 (0.61 to 0.83)	<.001	−0.87 (−1.89 to 0.08)	0.87 (0.75 to 1.02)	.24	−0.80 (−1.79 to 0.08)	0.88 (0.76 to 1.03)	.33
Quality of life	−1.74 (−2.66 to −0.90)	0.69 (0.57 to 0.82)	<.001	−0.71 (−1.59 to 0.06)	0.85 (0.71 to 1.02)	.24	−0.44 (−1.25 to 0.40)	0.90 (0.75 to 1.08)	.51
Physical health	−3.63 (−4.83 to −2.44)	0.66 (0.58 to 0.75)	<.001	−1.84 (−3.04 to −0.63)	0.80 (0.70 to 0.92)	.01	−1.96 (−3.15 to −0.86)	0.79 (0.69 to 0.90)	.004
Mental health	−1.49 (−2.72 to −0.32)	0.85 (0.75 to 0.97)	.02	0.31 (−0.89 to 1.39)	1.04 (0.91 to 1.18)	.76	0.09 (−1.02 to 1.18)	1.01 (0.89 to 1.15)	.93
Satisfaction with social activities	−1.92 (−3.25 to −0.65)	0.83 (0.74 to 0.94)	.004	−0.15 (−1.37 to 1.04)	0.98 (0.87 to 1.12)	.81	−0.06 (−1.23 to 1.16)	0.99 (0.88 to 1.13)	.93
Ability to carry out everyday physical activities	−0.31 (−0.79 to 0.08)	0.72 (0.48 to 1.09)	.12	−0.14 (−0.63 to 0.26)	0.86 (0.57 to 1.31)	.73	−0.07 (−0.51 to 0.28)	0.92 (0.61 to 1.40)	.90
Average fatigue in the past 7 d	−1.44 (−2.33 to −0.65)	0.72 (0.60 to 0.87)	<.001	−0.65 (−1.49 to 0.15)	0.86 (0.71 to 1.03)	.24	−0.64 (−1.49 to 0.16)	0.86 (0.71 to 1.03)	.33
Ability to carry out usual social roles	−0.87 (−1.92 to 0.15)	0.86 (0.72 to 1.01)	.07	0.14 (−0.78 to 1.01)	1.03 (0.86 to 1.22)	.81	0.27 (−0.59 to 1.17)	1.05 (0.89 to 1.25)	.84
Bothered by emotional problems past 7 d	−2.51 (−3.79 to −1.24)	0.79 (0.71 to 0.89)	<.001	−0.49 (−1.85 to 0.68)	0.95 (0.85 to 1.07)	.73	−0.66 (−1.82 to 0.58)	0.94 (0.83 to 1.06)	.51

Abbreviations: ADHD, attention-deficit/hyperactivity disorder; AOR, adjusted odds ratio; FDR, false discovery rate; pp, percentage points.

[a] Adjusted probability differences represent adult-diagnosed ADHD minus youth-diagnosed ADHD.

[b] AORs compare adult-diagnosed ADHD vs youth-diagnosed ADHD, with youth-diagnosed ADHD as the reference group.

[c] An FDR of less than .05 was applied across outcomes within each category (psychiatric comorbidities, somatic comorbidities, and overall health and daily activities), for a total of 9 tests per category. *P* values adjusted for FDR.

age, and income (eg, for bothered by emotional problems past 7 days: probability difference, −0.66 [95% CI, −1.82 to 0.58]) group differences were largely attenuated and were not significant (Table 2).

In sex-moderation analyses, a significant interaction was observed for autism spectrum disorder. Adult-diagnosed ADHD was associated with lower adjusted probabilities of autism compared with youth-diagnosed ADHD in both sexes, but this difference was stronger among males (interaction OR, 0.55 [95% CI, 0.38-0.80]; FDR-adjusted P = .02) (eTable 12 in Supplement 1).

In exploratory analyses separating youth-diagnosed ADHD into child- and adolescent-diagnosed groups, adult-diagnosed ADHD showed lower adjusted probabilities of most psychiatric comorbidities, somatic comorbidities, and impairment outcomes compared with both child- and adolescent-diagnosed ADHD. Differences between child- and adolescent-diagnosed ADHD were less consistent; adolescent-diagnosed ADHD was associated with lower adjusted probabilities for 3 of 9 psychiatric comorbidities and 1 of 9 somatic comorbidities, with no statistically significant differences in impairment outcomes (eTables 13-15 in Supplement 1).

Compared with their matched control participants without ADHD (eTable 2 and eTable 3 in Supplement 1), both the youth- and adult-diagnosed ADHD groups exhibited a markedly higher burden of all comorbid psychiatric conditions (eg, depression: probability difference, 39.33 [95% CI, 38.19-40.48] pp for adult-diagnosed ADHD and 36.92 [95% CI, 35.05-38.83] pp for youth-diagnosed ADHD) and somatic conditions (eg, brain and nervous system conditions: probability difference, 21.86 [95% CI, 20.64-22.93] pp for adult-diagnosed ADHD and 22.84 [95% CI, 21.05-24.83] pp for youth-diagnosed ADHD), except for cancer (probability difference, −0.42 [95% CI, −1.07 to 0.22] pp for adult-diagnosed ADHD and 0.05 [95% CI, −0.71 to 0.82] pp for youth-diagnosed ADHD) (**Table 3**; eTables 6-8 in Supplement 1). Both groups also showed greater functional impairment (eg, for bothered by emotional problems past 7 days: probability difference, 6.11 [95% CI, 5.45-6.80] pp for adult-diagnosed ADHD and 7.27 [95% CI, 5.99-8.60] pp for youth-diagnosed ADHD), with the exception of 1 measure (ability to carry out everyday physical activities: probability difference, 0.14 [95% CI, −0.05 to 0.39] pp for adult-diagnosed ADHD and −0.04 [95% CI, −0.42 to 0.39] pp for youth-diagnosed ADHD). Similar results were observed comparing with matched clinical controls (eTable 16 in Supplement 1).

Individuals with adult-diagnosed ADHD had lower adjusted ADHD-PRSs than those with youth-diagnosed ADHD (Cohen d, −0.17 [95% CI, −0.21 to −0.12]), and both groups had higher PRSs than their respective matched controls (adult-diagnosed ADHD vs matched controls mean difference Cohen d, 0.12 [95% CI, 0.09-0.15]; youth-diagnosed ADHD vs matched controls Cohen d, 0.21 [95% CI, 0.16-0.26]) (eTable 17 in Supplement 1). Across ADHD-PRS quintiles, higher quintiles were associated with lower odds of adult vs youth diagnosis (**Figure 2**A). For both youth- and adult-diagnosed groups, higher quintiles showed a stepwise increase in the odds of ADHD relative to matched controls (Figure 2B and C). Compared with matched clinical controls (eTable 4 and eTable 5 in Supplement 1), similar results were observed (eTables 9-11, eTable 18, and the eFigure in Supplement 1). Exploratory analyses separating youth-diagnosed ADHD into child- and adolescent-diagnosed groups showed that ADHD-PRSs were highest among child-diagnosed individuals, intermediate among adolescent-diagnosed individuals, and lowest among adult-diagnosed individuals (eTable 19 in Supplement 1).

Discussion

In this large All of Us sample, the following main conclusions were reached: (1) individuals first diagnosed with ADHD in adulthood had a milder clinical profile than individuals diagnosed in childhood or adolescence, represented by fewer psychiatric and somatic co-occurring conditions that are commonly observed in ADHD[17]; (2) adult-diagnosed individuals reported less impairment in overall health and daily activities than youth-diagnosed individuals; (3) relative to youth-diagnosed individuals, adult-diagnosed individuals had lower ADHD-PRSs, supporting an attenuated genetic profile of ADHD; (4) compared with matched controls without ADHD, both ADHD groups showed

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clinical and genetic profiles highly consistent with ADHD; and (5) compared with matched controls, both ADHD groups reported significant impairments in overall health and daily activities, indicating both as important subgroups warranting assessment and clinical management. Similar findings were observed using matched controls selected from a clinical sample. Taken together, our findings are most consistent with a severity continuum, rather than qualitatively distinct ADHD subtypes, defined by age at first diagnosis, with adult-diagnosed ADHD representing a comparatively milder expression of ADHD-like clinical and genetic profiles.

Our results showed lower rates of most psychiatric disorders in adult-diagnosed ADHD compared with youth-diagnosed ADHD, with depressive disorder as the only exception among the comorbidities assessed. These findings extend smaller clinic-based studies that have reported mixed

Table 3. Probability Differences and ORs for ADHD Groups Compared With Matched Controls

Diagnosis or impairment	Adult-diagnosed ADHD vs matched controls			Youth-diagnosed ADHD vs matched controls		
	Probability difference (95% CI), pp[a]	OR (95% CI)[b]	*P* value[c]	Probability difference (95% CI), pp[a]	OR (95% CI)[b]	*P* value[c]
Psychiatric comorbidities						
Anxiety disorder	34.74 (33.71 to 35.86)	4.65 (4.41 to 4.90)	<.001	35.96 (34.19 to 37.74)	4.58 (4.22 to 4.98)	<.001
Autism spectrum disorder	2.82 (2.46 to 3.19)	10.09 (7.67 to 13.28)	<.001	6.10 (5.29 to 6.94)	8.65 (6.59 to 11.36)	<.001
Bipolar disorder	9.26 (8.55 to 10.01)	3.84 (3.49 to 4.22)	<.001	14.46 (13.04 to 15.83)	4.64 (4.07 to 5.28)	<.001
Depression	39.33 (38.19 to 40.48)	5.44 (5.16 to 5.73)	<.001	36.92 (35.05 to 38.83)	4.79 (4.41 to 5.21)	<.001
Eating disorder	3.48 (3.04 to 3.93)	3.73 (3.20 to 4.35)	<.001	4.43 (3.62 to 5.26)	3.54 (2.87 to 4.37)	<.001
Personality disorder	3.17 (2.74 to 3.64)	3.74 (3.19 to 4.40)	<.001	7.50 (6.51 to 8.47)	4.90 (4.06 to 5.91)	<.001
Posttraumatic stress disorder	18.06 (17.13 to 19.06)	4.39 (4.09 to 4.72)	<.001	22.35 (20.87 to 23.86)	4.41 (3.99 to 4.88)	<.001
Schizophrenia	0.60 (0.40 to 0.84)	2.66 (1.96 to 3.63)	<.001	1.53 (1.08 to 2.06)	3.16 (2.25 to 4.43)	<.001
Substance use disorder	2.79 (2.34 to 3.30)	2.65 (2.30 to 3.06)	<.001	5.16 (4.24 to 6.07)	3.16 (2.63 to 3.79)	<.001
Somatic comorbidities						
Bone, joint, and muscle conditions	13.91 (12.86 to 15.12)	1.97 (1.87 to 2.08)	<.001	15.30 (13.59 to 16.86)	2.30 (2.10 to 2.51)	<.001
Brain and nervous system conditions	21.86 (20.64 to 22.93)	2.71 (2.57 to 2.85)	<.001	22.84 (21.05 to 24.83)	2.79 (2.57 to 3.03)	<.001
Cancer conditions	−0.42 (−1.07 to 0.22)	0.94 (0.86 to 1.03)	.20	0.05 (−0.71 to 0.82)	1.01 (0.84 to 1.22)	.90
Digestive conditions	12.94 (11.67 to 14.12)	1.81 (1.71 to 1.90)	<.001	16.56 (14.60 to 18.26)	2.18 (2.01 to 2.37)	<.001
Hearing and eye conditions	13.06 (11.79 to 14.26)	1.75 (1.66 to 1.84)	<.001	15.86 (13.86 to 17.84)	1.91 (1.76 to 2.06)	<.001
Heart and blood conditions	8.56 (7.33 to 9.72)	1.44 (1.37 to 1.52)	<.001	10.81 (8.89 to 12.61)	1.67 (1.54 to 1.82)	<.001
Hormone and endocrine conditions	7.07 (6.01 to 8.16)	1.43 (1.36 to 1.51)	<.001	6.90 (5.35 to 8.49)	1.52 (1.38 to 1.67)	<.001
Kidney conditions	1.76 (1.20 to 2.36)	1.42 (1.28 to 1.59)	<.001	1.60 (0.79 to 2.38)	1.44 (1.20 to 1.73)	<.001
Lung conditions	13.43 (12.39 to 14.51)	2.04 (1.93 to 2.16)	<.001	16.82 (15.09 to 18.39)	2.46 (2.25 to 2.69)	<.001
Overall health and daily activities						
General health	2.21 (1.66 to 2.75)	1.60 (1.43 to 1.78)	<.001	2.94 (1.97 to 3.97)	1.60 (1.37 to 1.86)	<.001
Quality of life	1.98 (1.53 to 2.44)	1.97 (1.71 to 2.27)	<.001	2.53 (1.75 to 3.39)	1.88 (1.55 to 2.27)	<.001
Physical health	2.72 (2.07 to 3.39)	1.56 (1.41 to 1.72)	<.001	4.42 (3.38 to 5.53)	1.71 (1.50 to 1.95)	<.001
Mental health	5.76 (5.14 to 6.38)	2.74 (2.48 to 3.04)	<.001	6.46 (5.33 to 7.66)	2.20 (1.92 to 2.51)	<.001
Satisfaction with social activities	5.98 (5.29 to 6.77)	2.33 (2.13 to 2.56)	<.001	5.53 (4.36 to 6.77)	1.90 (1.67 to 2.16)	<.001
Ability to carry out everyday physical activities	0.14 (−0.05 to 0.39)	1.20 (0.92 to 1.58)	.19	−0.04 (−0.42 to 0.39)	0.96 (0.65 to 1.42)	.83
Average fatigue in the past 7 d	2.06 (1.60 to 2.51)	2.13 (1.84 to 2.46)	<.001	2.67 (1.93 to 3.52)	2.04 (1.67 to 2.48)	<.001
Ability to carry out usual social roles	3.37 (2.87 to 3.88)	2.62 (2.30 to 2.98)	<.001	3.15 (2.31 to 4.11)	2.14 (1.77 to 2.58)	<.001
Bothered by emotional problems past 7 d	6.11 (5.45 to 6.80)	2.49 (2.26 to 2.73)	<.001	7.29 (5.99 to 8.60)	2.12 (1.88 to 2.40)	<.001

Abbreviations: ADHD, attention-deficit/hyperactivity disorder; FDR, false discovery rate; OR, odds ratio; pp, percentage points.

[a] Probability differences represent ADHD minus matched controls.

[b] ORs compare each ADHD group with its respective matched control group, with controls as the reference group. Models were unadjusted after propensity score matching.

[c] FDR less than .05 was applied separately for each ADHD group comparison within each outcome category (psychiatric comorbidities, somatic comorbidities, and overall health and daily activities), for a total of 9 tests per category per comparison. *P* values adjusted for FDR.

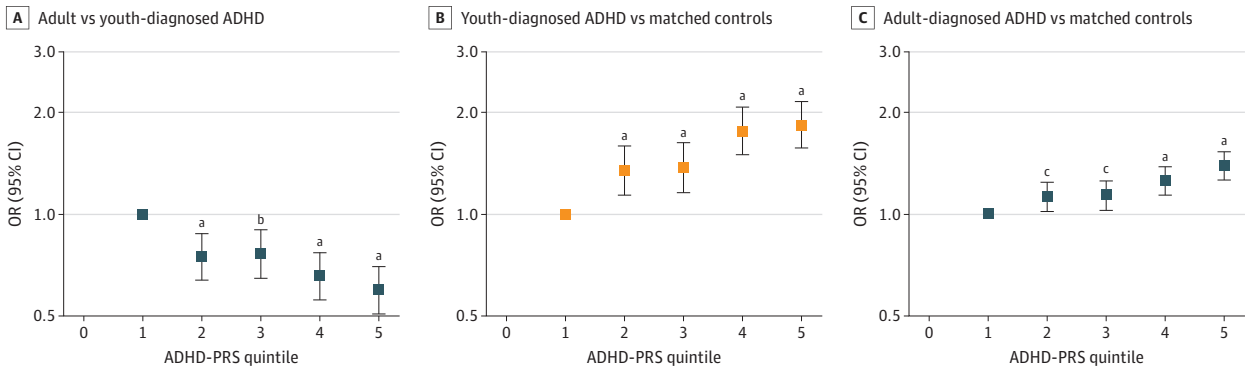
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results.[8,9] In one review of medical records at a university hospital, investigators compared 50 adults diagnosed with ADHD in childhood and 58 adults diagnosed in adulthood and found a higher total number of comorbidities in the childhood-diagnosed group.[8] A second study of 134 adults referred to an adult ADHD clinic, with age at diagnosis verified in electronic medical records, found no statistically significant differences between groups in prior exposure to antianxiety, mood stabilizer, or antipsychotic medications and no differences in quality of life assessed via structured questionnaires.[9] In a much larger and more diverse sample, our findings support higher rates of psychiatric comorbidities in youth-diagnosed ADHD relative to adult-diagnosed ADHD. Given that greater symptom severity and psychiatric comorbidity have been associated with persistence of ADHD into adulthood in longitudinal studies,[18] adults with youth-diagnosed ADHD who remain in clinical care may be expected to show greater psychiatric burden. Importantly, both ADHD groups had higher rates of every psychiatric comorbidity examined compared with their matched controls, consistent with prior literature showing elevated psychiatric burden in adults with ADHD.[19]

Our findings indicate that individuals with adult-diagnosed ADHD have lower rates of somatic comorbidities than those diagnosed with ADHD in youth. Both ADHD groups showed higher rates of somatic comorbidities than their matched controls, with cancer conditions as the only exception. This pattern aligns with prior work showing substantial somatic morbidity in ADHD populations, likely influenced by shared genetic vulnerability and overlapping risk factors.[20-22] In our data, the largest probability differences between ADHD groups and their matched controls were observed for neurological, respiratory, and musculoskeletal conditions, with differences of approximately 13 to 23 pp across comparisons, consistent with large epidemiological studies.[21] Prior studies also suggest that somatic comorbidities are more common with more severe ADHD presentations.[23,24] Taken together, these observations point to a higher somatic burden in youth-diagnosed ADHD compared with adult-diagnosed ADHD.

Regarding self-reported impairment, analyses adjusted for age and sex initially suggested lower impairment among adult-diagnosed than youth-diagnosed adults. After further adjustment for educational achievement or income, these differences were substantially attenuated and were not significant, indicating that socioeconomic status strongly influences the reported impairments. This is not surprising, given that socioeconomic factors are well-established determinants of overall health.[25] Moreover, higher educational achievement and income among adult-diagnosed individuals may partly reflect their older age and greater time to complete education and accumulate income. Both ADHD groups reported worse overall health and overall functional impairment in daily activities compared with their matched controls, in line with prior literature demonstrating functional

Figure 2. Dot Plots Illustrating Attention-Deficit/Hyperactivity Disorder (ADHD) Polygenic Risk Scores (PRS) by Age at Diagnosis



Odds ratios (ORs) and 95% CIs for ADHD across quintiles (Q2-Q5 vs Q1 [reference]) of the residualized ADHD-PRS, estimated with logistic regression models. A false discovery rate (FDR) adjustment was applied across the displayed Q2-Q5 quintile comparisons.

[a] FDR-corrected $P < .001$.

[b] FDR-corrected $P = .002$.

[c] FDR-corrected $P = .01$.

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impairment in ADHD.[26,27] Thus, despite fewer comorbidities, individuals diagnosed with ADHD as adults report a similar level of functional impairment once socioeconomic status is accounted for, underscoring the importance of recognizing and treating ADHD even when diagnosis occurs in adulthood.

We observed that individuals in the adult-diagnosed ADHD group had lower ADHD-PRSs than those in the youth-diagnosed ADHD group. Consistent with our findings, LaBianca et al[28] observed higher ADHD-PRSs among individuals first diagnosed in childhood than among those first diagnosed in adulthood. In contrast, Rajagopal et al[29] found no statistically significant differences after correction for multiple comparisons, with only nominal differences between individuals diagnosed in childhood and adulthood vs those diagnosed only in adulthood. Notably, both studies showed clear ADHD-PRS elevations in adult-diagnosed cases relative to controls without ADHD.[28,29] This finding aligns with our results, in which both ADHD groups had higher ADHD-PRSs than their matched controls. The effect sizes for ADHD-PRS vs controls in our study are at the lower end of those reported in the literature,[30-32] which may reflect the use of self-reported ADHD diagnosis, introducing misclassification that can attenuate genetic findings. Overall, ADHD-PRS patterns are consistent with adult-diagnosed ADHD representing a less severe expression of ADHD liability.

Most individuals with ADHD in our sample were women in both the adult- and youth-diagnosed groups. This contrasts with the broader literature, in which ADHD is more prevalent among boys in childhood and roughly comparable by sex in adulthood.[33] Our definition of ADHD required currently seeing a clinician, which may bias toward female participants, as women are more likely to seek professional help for mental health concerns.[34] Although women have been reported to seek an ADHD evaluation in adulthood more often than men,[35] we did not observe a higher proportion of women in the adult-diagnosed group compared with the youth-diagnosed group. This may reflect the high prevalence of women in both groups, which could dilute between-group differences. Exploratory moderation analyses did not suggest broad sex differences in the associations between adult- vs youth-diagnosed ADHD. Autism spectrum disorder was the only outcome with a statistically significant interaction after correction for multiple comparisons, with adult-diagnosed ADHD showing lower probability of autism spectrum disorder in both sexes but a larger difference among males. This may reflect sex differences in neurodevelopmental diagnostic recognition, whereby co-occurring autism spectrum disorder may facilitate earlier ADHD identification more strongly among males than females.

Exploratory analyses separating child- and adolescent-diagnosed ADHD were broadly consistent with a graded pattern in which earlier diagnosis is associated with greater genetic and clinical burden, particularly psychiatric comorbidity. However, this pattern was not uniform across clinical domains and may have been blurred by retrospective misclassification of childhood vs adolescent diagnosis timing.

Notably, this study did not evaluate the late-onset ADHD hypothesis.[36] That is, we were unable to determine absence of childhood symptoms, which would be required to substantiate late-onset ADHD, since childhood symptom data were unavailable. Our goal was conceptually different and focused on characterizing the growing number of individuals receiving a first ADHD diagnosis in adulthood, irrespective of childhood symptom history. Our findings should not be interpreted as evidence for or against late-onset ADHD.

Limitations

Findings from this study should be interpreted with consideration of the following limitations. First, this study relied on self-reported ADHD diagnoses, including age at first diagnosis, and on self-reported co-occurring conditions, which may be subject to misclassification or recall bias. Importantly, the ADHD-PRSs showed patterns across groups that were consistent with the clinical results, supporting the overall validity of the classification. Second, despite propensity score matching, residual confounding is possible, including differences in access to care, regional practice patterns, and screening intensity. Third, selection bias inherent to volunteer research cohorts may

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limit generalizability. The All of Us research Program deliberately prioritizes groups historically underrepresented in biomedical research[10] but is not a probability-based sample of the US population. In the present analysis, 4.5% of the assessed sample met our ADHD definition. This estimate should not be interpreted as population prevalence, but it is broadly compatible with recent nationally representative Centers for Disease Control and Prevention data showing that 6.0% of US adults reported current ADHD.[2] However, the sociodemographic composition of our ADHD analytic sample differed from nationally representative estimates of adults with current ADHD, including a higher proportion of female participants, a lower proportion of non-Hispanic Black participants, and substantially higher educational attainment.[2] Together, these differences suggest that the present results should be interpreted as comparisons among All of Us participants meeting the study definitions rather than as nationally representative estimates of ADHD frequency, comorbidity, or impairment. Fourth, our ADHD definition required participants to report currently seeing a clinician for ADHD, which limits generalizability. Adults with greater health care access or health-seeking behavior may be more likely to obtain a diagnosis, and those with more severe symptoms may be more likely to remain in care. Fifth, our aim was to compare adults with current ADHD by age at diagnosis; thus, individuals diagnosed in youth who are not in clinical care as adults (and are more likely to be remitted) were outside our scope. Sixth, controls without ADHD included individuals who reported no lifetime ADHD diagnosis; however, some individuals with undiagnosed ADHD may have been included in this group.

Conclusions

In this cross-sectional study of adults with ADHD, adults first diagnosed with ADHD in adulthood showed a milder clinical and genetic profile than individuals first diagnosed in childhood or adolescence, which may have contributed to later recognition. However, both groups had clinical profiles typical of individuals with ADHD relative to individuals without ADHD, with substantial burden. These findings support the premise that adult-diagnosed ADHD is a clinically important subgroup that warrants recognition and treatment.

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SUPPLEMENT 1.

eMethods.

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SUPPLEMENT 2.
Data Sharing Statement

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